What Reveals Below: Fold-Contact Affordances for Cloth Unfolding

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Abstract: Unfolding a crumpled garment from an overhead view is ambiguous: a flat cloth and a stack of hidden folds can share the same silhouette. This paper studies a transparent-table dual-arm setup in which the robot observes both a top segmentation mask and a bottom contact mask. The main finding is that the bottom view is most useful when converted into fold-contact affordances rather than used as a generic second image. We derive fold residual, contact overlap, fold-contact frontier, and silhouette-edge maps from the dual masks, use them in an attention encoder, and further convert the frontier into a residual action prior. To reduce hand tuning, we parameterize the prior's cell scorer as an online linear, contextual-bandit-style update trained from reward, self-supervised unfolding progress, or their mixture. Across the completed logs, raw dual-view pixels improve long-horizon flatness over a top-only policy, affordance channels reduce folded area by 82.2% relative to raw dual masks in a fast screen, and a reward-bandit bottom-view scorer gives the best 21-run residual-prior ablation with within-panel geometry score 0.840 ± 0.047. The results support a precise conclusion: bottom view helps cloth unfolding by exposing where upper folded layers meet grounded support, and reward-trained scoring is the most effective tested way to turn that information into manipulation targets.

Keywords: Cloth unfolding, Bottom-view sensing, Affordances

1 Introduction

Robotic cloth unfolding is a useful testbed for deformable-object manipulation because it asks a robot to reason about shape, contact, and long-horizon action effects from incomplete visual evidence. A crumpled garment can hide large parts of its state under self-occluding layers, and the same top-view silhouette can correspond to very different physical configurations: a nearly flat cloth, a stack of folds, or a lifted flap that has not yet made contact with the table. This ambiguity is not a minor perceptual nuisance. In a dual-arm unfolding policy, the useful action often depends on which side of a fold is grounded and which side should be pulled away; top-view area alone does not reveal that support relationship.

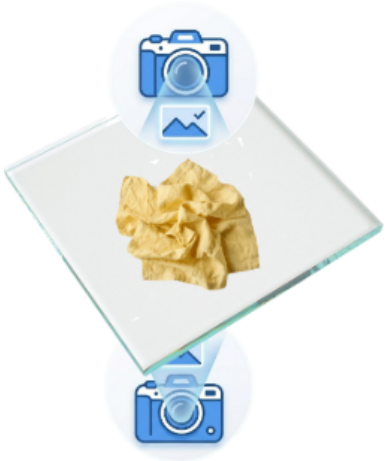


Figure 1: Top view and bottom view.

The ambiguity also makes reinforcement learning (RL) difficult to interpret and optimize. A policy can increase projected area while leaving hidden folded layers stacked, or it can move cloth through visually plausible silhouettes without improving grounded contact. Binary unfolding success is also threshold-sensitive: it depends on garment category, initialization, and

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the chosen tolerance for coverage and flatness. Like much prior work in cloth manipulation, we therefore use continuous geometry progress measures to diagnose what improves during unfolding. For this reason, the central question in this paper is not only how to add more visual input, but how to expose the particular contact information that is missing from the overhead observation.

We study a transparent-table setting in which the robot observes both a top-view cloth silhouette and a bottom-view contact mask, or the corresponding simulated masks. The bottom view reveals which occupied cells are near the table surface, making it possible to distinguish grounded support from cloth that is visible from above but suspended or folded over another layer. Directly feeding top-plus-bottom pixels to a policy is a natural baseline, but our experiments show that the larger gain comes from converting the two masks into structured fold-contact cues: residual folded area, grounded contact, the frontier between folded and supported regions, and the outer silhouette edge.

These cues are useful in two complementary ways. First, they form an observation representation that gives the policy explicit access to the hidden support structure behind a top silhouette. Second, the fold-contact frontier is an actionable affordance: it marks candidate grasp locations where a dual-arm pull can separate an upper folded layer from grounded cloth. We therefore use the frontier both inside a contact-affordance attention encoder and as the source of a residual action prior. The learned SAC actor still chooses continuous corrections, but the prior biases exploration toward bottom-view targets that are geometrically meaningful.

The remaining design question is how to rank frontier targets. A fixed geometric scorer is interpretable but brittle because the best pick depends on the current fold, contact, and pull direction. We replace hand-tuned frontier weights with a lightweight online linear scorer updated in a contextual-bandit-style manner from reward, self-supervised unfolding progress, or a mixture of the two. This scorer is not meant to replace policy learning; it adapts the action prior's target selection while the residual RL policy handles continuous control.

This paper makes three contributions grounded in completed experiments:

1. A sensor-motivated fold-contact observation model that converts top and bottom masks into residual fold, contact, frontier, and edge maps, exposing support information that is hidden in the overhead silhouette.
2. A bottom-view residual action prior that selects fold-contact frontier targets while leaving SAC to learn continuous residual corrections for dual-arm motion.
3. A learned affordance scorer that replaces fixed frontier weights with an online bandit-style update. The best variant learns from environment reward, raises the within-ablation geometry score from 0.592 to 0.840 over a fixed heuristic residual prior, and reduces folded area by 65.4% while keeping area nearly unchanged.

2 Related Work

Robotic cloth manipulation is difficult because cloth has high-dimensional state, self-contact, and partial observability; surveys emphasize that methods often succeed by exploiting task structure in perception, modeling, or control [1, 2]. Early systems used geometric cues and structured manipulation sequences for towel folding or clothing configuration estimation [3, 4], while benchmarks for bimanual cloth manipulation formalized tasks such as spreading, folding, and dressing with task-specific quality measures [5]. Learning-based approaches have since used imitation, scripted supervisors, and sim-to-real training to reduce manual policy design [6, 7, 8]. These works motivate our choice to expose physically meaningful variables instead of asking a policy to infer all cloth structure from pixels.

Recent cloth-unfolding and smoothing methods show the value of strong visual and action priors. Fabric smoothing has been learned from algorithmic supervision [8]; dynamic dual-arm flinging can rapidly increase coverage [9]; flow-based bimanual policies use visual correspondence as both an

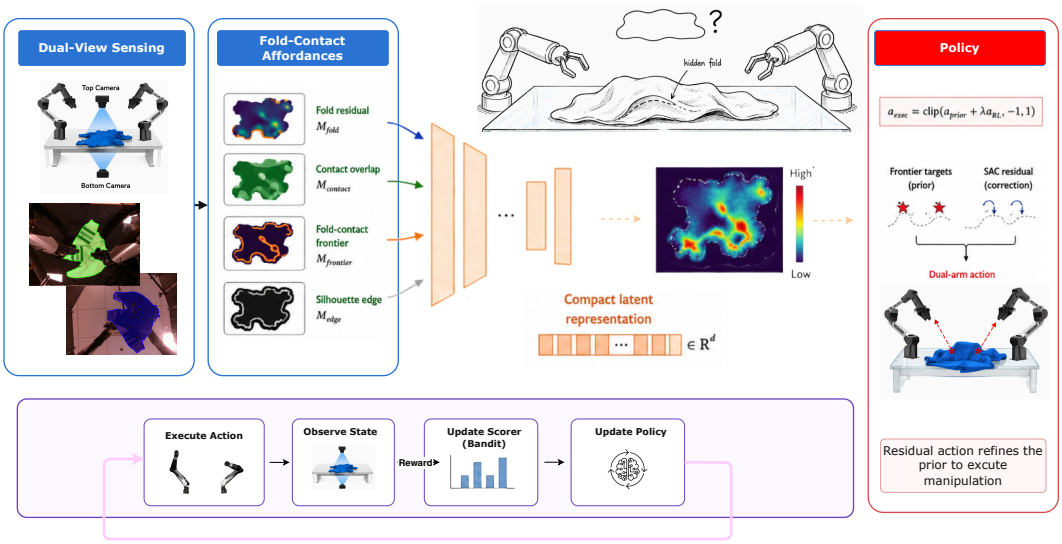


Figure 2: Method pipeline. **(1)** Dual-view sensing records a top silhouette and a bottom contact mask. **(2)** The two masks are converted into fold-contact affordances: fold residual, contact overlap, fold-contact frontier, and silhouette edge. **(3)** A multi-channel affordance encoder maps these cues into a compact latent representation. **(4)** A learned bandit-style scorer ranks frontier cells as candidate manipulation targets. **(5)** The policy combines the selected frontier prior with SAC residual actions. The lower loop summarizes online interaction: execute an action, observe the next cloth state, compute reward, update the frontier scorer, and update the SAC actor–critic from experience.

observation and action representation [10]; and efficient folding systems combine learned bimanual primitives with downstream folding routines [11]. Other work changes the actuation itself, for example using airflow to apply dense forces during deformable-object manipulation [12]. Our setting keeps contact-based dual-arm pulling, but changes the sensing assumption: a transparent table gives a bottom contact mask in addition to the top silhouette.

Partial observability is a central obstacle for cloth state estimation. SoftGym provides deformable-object simulation tasks for learning-based manipulation [13], while visible-connectivity dynamics infer graph structure from partial point clouds for cloth smoothing [14]. GarmentNets addresses category-level garment pose estimation through canonical-space shape completion [15], and MEDOR explicitly reasons about occluded cloth regions by reconstructing a mesh and refining it with self-supervised test-time losses [16]. In contrast, we do not try to recover a full 3D cloth state. We use the bottom mask to expose a narrower support relation: which top-visible cells are grounded and where folded material meets contact.

Deep RL algorithms such as SAC, TD3, and PPO provide general continuous-control baselines [17, 18, 19], and affordance-oriented manipulation shows that action-relevant spatial structure can improve exploration and control [20, 21]. Transporter networks further show that spatial action parameterizations and visual correspondences can make manipulation policies more sample efficient [22]. For cloth unfolding, the main difficulty is not only action dimensionality but also misleading top-view progress: projected area can increase while hidden folds remain stacked. We therefore use bottom-view fold-contact frontiers both as observation channels and as residual action-prior targets, leaving SAC to learn continuous corrections rather than replacing the policy with a scripted controller.

The learned frontier scorer is related to contextual bandits and lightweight online adaptation, where action preferences are updated from observed feedback and UCB-style bonuses encourage uncertain choices [23, 24]. We use these ideas narrowly: a linear scorer ranks local frontier cells from fold/contact/edge features and is updated from bounded reward or self-supervised progress. It does

not model long-horizon cloth dynamics or claim a regret guarantee; its purpose is to adapt the target-selection part of an interpretable bottom-view action prior.

3 Method

Task. The simulated task uses two end effectors and a crumpled short-sleeve garment on a table. At each control step the action is

$$a_t = [\Delta x^L, \Delta y^L, \Delta z^L, \Delta\theta^L, \Delta g^L, \Delta x^R, \Delta y^R, \Delta z^R, \Delta\theta^R, \Delta g^R].$$

The proprioceptive state contains end-effector poses, target poses, gripper openings, and joint positions. The reward is grounded: area, spread, and flatness gains are gated by cloth height above the table, with penalties for action magnitude, grasp stalls, and pulling without cloth progress.

Dual-view masks. Let M_t^{top} be the top-view cloth silhouette over a fixed table-centered grid. In simulation, the bottom mask is computed from cloth particles and table height; it is therefore privileged relative to a real upward-facing camera. The mask is active where the lowest cloth surface in an occupied cell lies near the table:

$$M_t^{bot}(i,j) = \mathbb{1}\left[M_t^{top}(i,j) = 1 \wedge z_{i,j}^{min} \le z_{table} + \epsilon_{bot}\right].$$

The top-only baseline observes $[s_t^p, M_t^{top}]$, while the raw dual-view baseline observes $[s_t^p, M_t^{top}, M_t^{bot}]$. We add mask dropout and random flips during training, and the occlusion stress test masks the policy observation; unmodeled real effects such as calibration drift, reflections, and segmentation bias remain limitations.

Fold-contact affordance channels. The affordance observation explicitly separates upper folded cloth from grounded support:

$$M_t^{fold} = \max(M_t^{top} - M_t^{bot}, 0), \qquad M_t^{contact} = M_t^{top} M_t^{bot},$$

$$M_t^{frontier} = \mathrm{dilate}(M_t^{fold})\,\mathrm{dilate}(M_t^{contact}) M_t^{top}, \qquad M_t^{edge} = M_t^{top} - \mathrm{erode}(M_t^{top}).$$

The implementation uses one iteration of a 3×3 square binary kernel for each dilation/erosion operation, a 32×32 mask grid, and $\epsilon_{bot} = 3.5$ cm unless otherwise stated. The policy receives $[s_t^p, M_t^{top}, M_t^{bot}, M_t^{fold}, M_t^{contact}, M_t^{frontier}, M_t^{edge}]$. A contact-affordance attention encoder applies separate visual stems to fold/frontier and contact/edge channels, exchanges tokens with bidirectional attention, pools them, and concatenates the result with proprioception before the actor and critic. Appendix D gives the exact channel widths, tokenization, and checkpoint-selection protocol.

Residual bottom-view action prior. The fold-contact frontier also defines an action prior. For each arm, candidate cells are restricted to the corresponding side of the garment. A cell feature vector

$$\phi_t(i,j) = [M^{frontier}, M^{edge}, M^{fold}, M^{top}, M^{contact}, d^{out}]$$

is scored, the best cell becomes a pick target, and the prior moves the gripper toward the target, closes near it, and pulls outward from the garment center. The executed action is residual:

$$a_t^{exec} = \mathrm{clip}\Big(a_t^{RL} + \lambda_t a_t^{prior}, -1, 1\Big).$$

Learning the scorer. The fixed prior uses hand-chosen weights over ϕ. We replace them with a linear, contextual-bandit-style scorer $s_t(i,j) = w_t^\top \phi_t(i,j)$. For each arm $r \in \{L, R\}$, the candidate set $\mathcal{C}_t^r$ contains top-mask cells on that arm's side of the table-centered ROI with finite score; if all learned scores are non-positive, the implementation falls back to a top-mask plus outward-distance score on the same side. The selected cell $u_t^r = (i_t^r, j_t^r)$ is greedy except for ϵ-exploration, which samples uniformly from the top quartile of finite-scored candidates on that side. At each policy step we form bounded feedback $p_t \in [-1, 1]$ from either grounded geometry progress g_t, reward $\tanh(R_t/2)$, or their mixture. With EMA baseline $\bar{a}_t$, clipped advantage $A_t = \mathrm{clip}(p_t - \bar{a}_t, -1, 1)$,

and centered feature delta $\Delta\phi_t$ between selected frontier cells and their candidate-set mean, the weights are updated by

$$w_{t+1} = \Pi_{[l,u]}\big[w_t + \eta A_t\, \Delta\phi_t + \eta\rho(w_0 - w_t)\big],$$

where $\Pi_{[l,u]}$ clips each coordinate. The bounded feedback, centered features, decay toward the heuristic w_0, and projection keep each update bounded; we use η=0.20, ρ=0.01, β=0.90, and ϵ=0.10 in the reward-bandit runs. This is best viewed as a projected advantage update for target ranking, not as a bandit algorithm with a regret guarantee. Appendix N.1 gives the exact pseudo-code, stability argument, and UCB-style uncertainty variant.

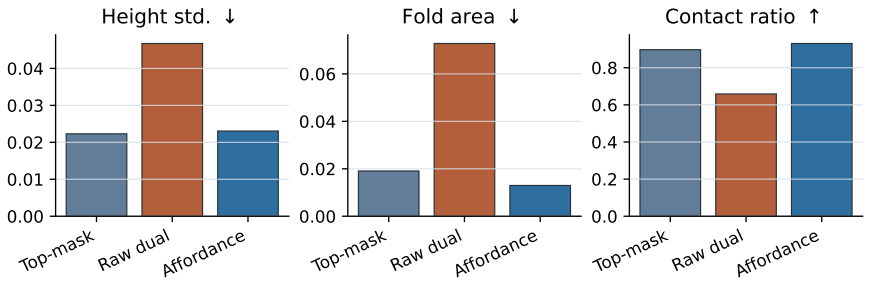


Figure 3: Bottom-view impact from existing logs. Raw dual view improves some flatness diagnostics over top-only observation, but the fold-contact affordance decomposition is what sharply reduces folded area and recovers grounded contact in the fast screen. The top-mask fold/contact bars are filled from the current-metric top-only evaluation because the earliest one-million-step top run did not log these bottom-derived diagnostics.

4 Experiments

Metrics. For cloth unfolding, a binary success rate is highly dependent on the garment, initial crumple, and chosen coverage or flatness threshold. We therefore follow the common practice of reporting continuous progress metrics: evaluation return, final cloth area A for coverage, folded mask area F for residual folds, contact ratio C for grounded support, and height standard deviation Z for flatness. For screening we use

$$S = 0.35A + 0.25C + 0.25(1 - F) + 0.15(1 - Z),$$

where each component is min-max normalized only within the variants in the same table or figure; higher area/contact and lower fold/z-std are better. We use S only as a within-panel ranking index; cross-block comparisons should use the absolute return, area, fold, contact, and z-std columns.

Validation scope. Several appendix studies support the main experiments without changing the central ranking criterion. Qualitative rollouts from the same crumpled state make the principal failure mode visible: the top-only policy can increase projected area while leaving a folded front layer unresolved, whereas the reward-bandit affordance policy keeps both arms engaged and reduces the final folded residue from $F \approx 0.017$ to $F \approx 0.004$ with contact near 0.98. Mid-budget validation runs test whether the short scorer ablation is a pure horizon artifact: a 75k-step reward-bandit versus heuristic-prior comparison preserves the scorer advantage with a 0.168 within-table score gain, -0.0021 z-std change, and -0.0033 folded-area change. The offline mask-sensitivity probe further shows that threshold and morphology choices are relatively stable, while calibration-style shifts are the dominant failure: a one-pixel bottom-mask shift drops frontier IoU to 0.065 and moves picks by 23.2 cm. These diagnostics motivate the robustness tests below and are reported in full in Appendices B, J, and K.

4.1 Does Bottom View Add Information?

The one-million-step comparison shows the first effect of bottom view. Raw dual-view cross-attention only slightly changes return and area relative to the top-mask baseline, but it lowers height standard

deviation from 0.0271 to 0.0218, a 19.8% reduction. This confirms that bottom-view contact cues help flatness, while also showing that raw pixels alone are not enough to produce a strong unfolding method, as summarized in Figure 3.

The fast screen isolates the representation. Raw dual masks obtain high area but leave substantial folds: folded area 0.0728, contact ratio 0.659, and z-std 0.0467. The affordance observation lowers folded area to 0.0130 and z-std to 0.0231, while increasing contact ratio to 0.931. Thus the bottom view helps most when interpreted as support/contact structure rather than as an unstructured second image.

4.2 Can Bottom View Drive Better Actions?

A no-learning 80-step action-prior probe shows that bottom-view frontiers are actionable: the affordance prior increases area by 0.0469 versus 0.0072 for zero action and 0.0052 for random dual actions. However, the pure prior can slightly hurt contact and fold quality. Longer residual-policy training confirms the trade-off: a 12-hour residual-prior run improves the geometry score over affordance observation alone (0.650 versus 0.350), mainly through lower fold, higher contact, and lower z-std, but sacrifices some area and return. This motivates learning how to rank frontier targets instead of relying on fixed heuristics.

4.3 Which Scorer Is Best?

Table 1 and Figure 4 summarize the 21-run ablation. The reward-bandit scorer is the best variant, with geometry score 0.840 ± 0.047, return 33.4 ± 1.9, folded area 0.0068 ± 0.0035, contact ratio 0.964 ± 0.018, and z-std 0.0173 ± 0.0013. Compared with the heuristic residual prior, it raises the within-ablation score from 0.592 to 0.840 and reduces folded area by 65.4% while keeping area nearly unchanged.

Table 1: Learned bottom-view scorer ablation. Score uses area/contact positively and fold/z-std negatively; values are mean $\pm$ standard error over three seeds.

Variant	Runs	Score	Eval return	Area	Fold ↓	Contact ↑	Z-std ↓
Reward-bandit scorer	3	0.840 ± 0.047	33.4 ± 1.9	0.213 ± 0.004	0.0068 ± 0.0035	0.964 ± 0.018	0.0173 ± 0.0013
Progress scorer + adaptive prior	3	0.823 ± 0.010	30.7 ± 0.7	0.213 ± 0.002	0.0072 ± 0.0002	0.963 ± 0.002	0.0181 ± 0.0004
Hybrid progress/reward scorer	3	0.775 ± 0.119	27.1 ± 4.8	0.203 ± 0.012	0.0054 ± 0.0018	0.968 ± 0.013	0.0209 ± 0.0035
Hybrid scorer + adaptive prior	3	0.730 ± 0.179	30.9 ± 6.5	0.200 ± 0.022	0.0064 ± 0.0019	0.958 ± 0.017	0.0218 ± 0.0048
Reward scorer + adaptive prior	3	0.712 ± 0.110	32.3 ± 3.0	0.213 ± 0.011	0.0121 ± 0.0054	0.934 ± 0.027	0.0226 ± 0.0029
Progress-bandit scorer	3	0.706 ± 0.091	29.2 ± 0.7	0.218 ± 0.006	0.0139 ± 0.0058	0.927 ± 0.030	0.0238 ± 0.0041
Heuristic residual prior	3	0.592 ± 0.159	-4.6 ± 31.8	0.216 ± 0.006	0.0197 ± 0.0101	0.898 ± 0.049	0.0265 ± 0.0059

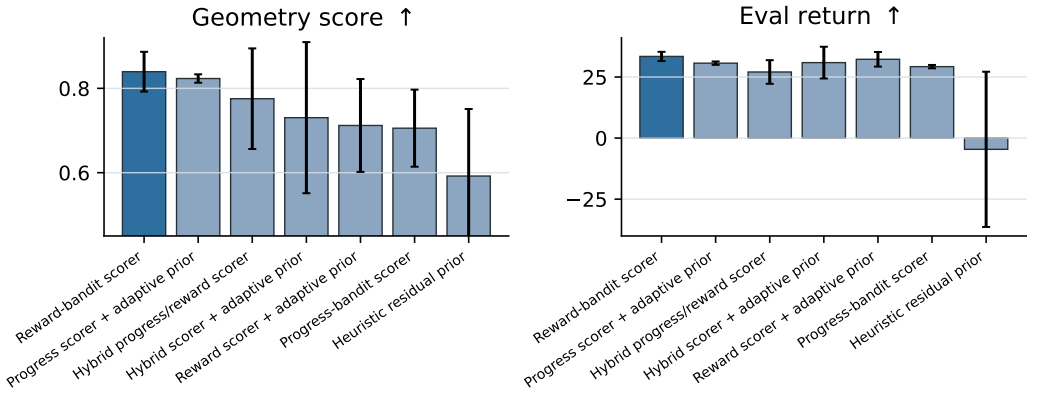


Figure 4: Algorithmic enhancement from the 21-run ablation. The reward-bandit scorer is the best overall method; adaptive scaling mainly helps the progress-only scorer.

The adaptive gate is useful for progress-only self-supervision: adding it raises the progress scorer from score 0.706 to 0.823 and stabilizes the three seeds. It does not improve the reward scorer,

suggesting that reward already penalizes the late over-pulling behavior that the gate was designed to reduce. The hybrid progress/reward scorer gives strong fold/contact in two seeds but has larger variance and lower mean area, so it is not the recommended method. Qualitative trajectory snapshots of the top-only baseline and the reward-bandit affordance policy starting from the same crumpled cloth state are provided in Appendix B (Figure 8).

We also test a UCB-style reward scorer that adds a count-based exploration bonus to the same linear score. It is statistically tied with reward-bandit on coverage and z-std at 25k steps, with slightly lower folded area and no collapsed seed in this small comparison. The full table and discussion are in Appendix M.

4.4 Robustness Diagnostics

We keep robustness claims secondary, but test whether the bottom-view cues survive two likely failure modes. Under arm-like top-camera occlusion, the reward-bandit affordance policy remains stronger than raw dual-view SAC, raising the within-panel geometry score by 0.152 and return by 16.6 because the bottom mask supplies information missing from the occluded top silhouette.

Second, we examine the proposed flatness mechanism. Across the 21-run scorer ablation, lower folded area is associated with lower height standard deviation ($r = 0.78$), while higher contact ratio is associated with lower height standard deviation ($r = -0.85$). This supports the interpretation that bottom-view affordances improve flatness by reducing folded residuals and increasing grounded contact, rather than merely increasing projected area.

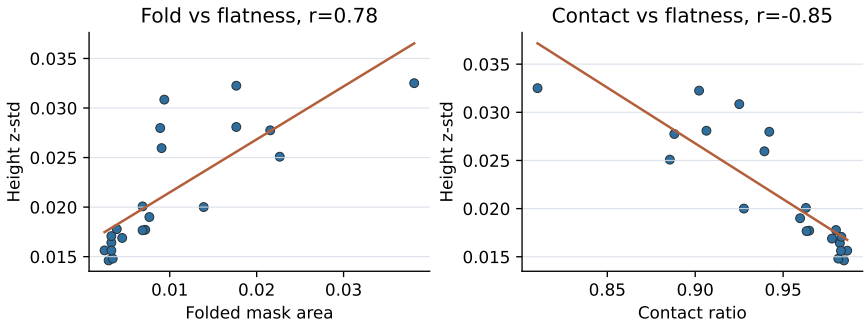


Figure 5: Mechanism behind the flatness improvement. Across the 21-run scorer ablation, lower folded area is positively associated with lower height standard deviation ($r = 0.78$), while higher contact ratio is negatively associated with height standard deviation ($r = -0.85$).

Bottom-mask noise is more delicate. False negatives and random flips are mild: the clean-trained reward-bandit score is 0.656 under FN 5% and 0.671 under flip 2%, compared with 0.653 on clean input. A one-pixel calibration shift drops the score to 0.366, reduces return from 32.8 to -409.7, and roughly triples folded area to 0.029; mixed failure behaves similarly. Training with mixed bottom-mask noise closes most of that gap without hurting clean evaluation, but these results still depend on a synthetic perturbation family. Appendix I contains the full occlusion, mask-sensitivity, and policy-level noise results.

4.5 Real-World Evaluation

We also run a representative real-robot trial on the dual-arm transparent-table setup to check whether the perception-driven progress measures transfer beyond simulation. The system records synchronized RGB observations, robot states, segmentation outputs, and reward-related metrics. The trial shows increasing progress metrics as the robot changes the garment configuration, suggesting that the same feedback pipeline can track physical manipulation progress despite sensor noise and cloth deformation. Full records and aligned keyframe/metric plots are reported in Appendix C.

Table 2: Policy-level bottom-mask noise robustness. Metrics use clean simulator geometry; only the policy's bottom-mask input is perturbed. "Score" is the within-panel composite S; absolute fold/contact/z-std columns should be used when comparing across rows.

Policy	Bottom-mask noise	Score	Return	Area	Fold ↓	Contact ↑	Z-std ↓
Top-only SAC	Clean	0.826 ± 0.028	29.4 ± 5.9	0.213 ± 0.009	0.0025 ± 0.0006	0.986 ± 0.003	0.0212 ± 0.0005
Reward-bandit affordance (clean-trained)	Clean	0.765 ± 0.035	32.8 ± 3.3	0.218 ± 0.005	0.0109 ± 0.0037	0.944 ± 0.018	0.0220 ± 0.0018
Reward-bandit affordance (clean-trained)	FN 5%	0.770 ± 0.016	30.9 ± 2.5	0.214 ± 0.004	0.0122 ± 0.0011	0.935 ± 0.005	0.0169 ± 0.0008
Reward-bandit affordance (clean-trained)	Flip 2%	0.780 ± 0.023	32.0 ± 3.6	0.215 ± 0.006	0.0106 ± 0.0022	0.945 ± 0.011	0.0183 ± 0.0011
Reward-bandit affordance (clean-trained)	1-px shift	0.563 ± 0.023	-409.7 ± 241.5	0.203 ± 0.005	0.0291 ± 0.0015	0.833 ± 0.008	0.0220 ± 0.0013
Reward-bandit affordance (clean-trained)	Mixed	0.554 ± 0.016	-283.1 ± 260.4	0.221 ± 0.004	0.0383 ± 0.0018	0.800 ± 0.008	0.0212 ± 0.0014
Reward-bandit affordance (noise-trained)	Clean	0.730 ± 0.029	22.7 ± 2.5	0.196 ± 0.004	0.0077 ± 0.0017	0.953 ± 0.011	0.0210 ± 0.0014
Reward-bandit affordance (noise-trained)	FN 5%	0.616 ± 0.040	21.2 ± 3.5	0.189 ± 0.006	0.0164 ± 0.0017	0.894 ± 0.014	0.0234 ± 0.0021
Reward-bandit affordance (noise-trained)	Flip 2%	0.670 ± 0.039	20.5 ± 3.3	0.192 ± 0.007	0.0119 ± 0.0017	0.924 ± 0.012	0.0229 ± 0.0017
Reward-bandit affordance (noise-trained)	1-px shift	0.451 ± 0.030	-2.4 ± 18.9	0.177 ± 0.009	0.0283 ± 0.0028	0.815 ± 0.014	0.0275 ± 0.0016
Reward-bandit affordance (noise-trained)	Mixed	0.410 ± 0.028	21.9 ± 3.6	0.183 ± 0.008	0.0363 ± 0.0019	0.768 ± 0.009	0.0255 ± 0.0016

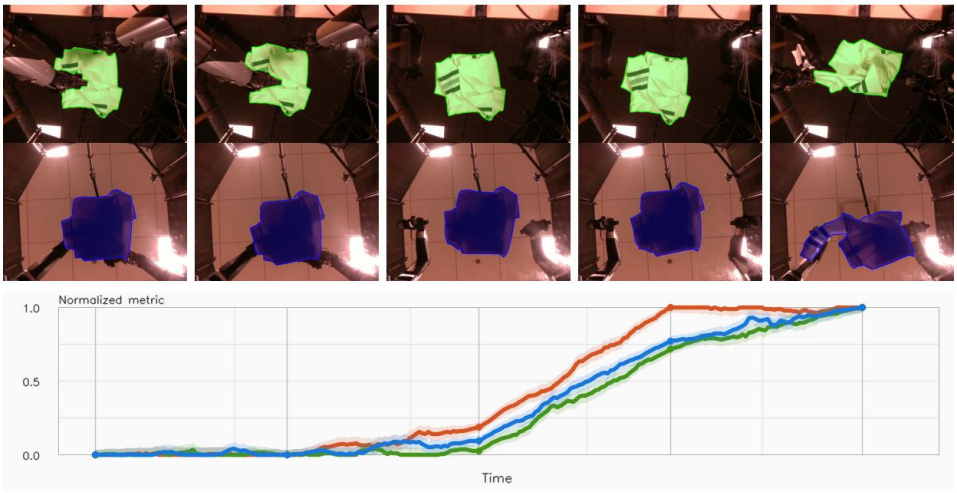


Figure 6: Real-world garment manipulation trial with perception-driven progress metrics. The top panel shows representative keyframes from a real robot execution, with the garment state segmented by SAM 3 [25] from synchronized camera observations. The bottom panel plots normalized progress metrics: cloth height decrease, upward lifting motion, and outward pulling alignment. The curves are extracted from the same trial and aligned with the displayed keyframes; their increasing values indicate that the policy progressively improves the garment configuration during physical interaction.

5 Limitations & Conclusion

The completed experiments support a focused claim: bottom view is valuable for cloth unfolding because it reveals the contact structure hidden by a top silhouette. Converting that structure into fold-contact affordances gives much better fold/contact/flatness diagnostics than raw dual masks. Turning the same frontiers into a residual action prior is useful, but fixed geometric weights are brittle; the strongest method in the current data is a reward-trained contextual-bandit scorer for bottom-view frontier targets.

Several limitations remain in the current real-world setup. The reachable workspace of the present dual-arm system is limited, so we use a simple left/right workspace partition with a lightweight obstacle-avoidance heuristic, allowing some overlap between the two arms. As a result, some desired manipulation points cannot be reached exactly in physical trials. Future work will evaluate the same perception and affordance pipeline on dual-arm platforms with larger and better conditioned reachable workspaces, which may further improve execution quality. The current real-robot experiments also still require a human operator to reset the garment between trials. Finally, because the system relies on depth sensing from both views, it is sensitive to external illumination and other optical disturbances; in practice, the experimental area must be enclosed with blackout curtains to obtain stable observations.

References

[1] J. Sanchez, J.-A. Corrales, B.-C. Bouzgarrou, and Y. Mezouar. Robotic manipulation and sensing of deformable objects in domestic and industrial applications: A survey. *The International Journal of Robotics Research*, 37(7):688–716, 2018. doi:10.1177/0278364918779698.

[2] A. Longhini, Y. Wang, I. Garcia-Camacho, D. Blanco-Mulero, M. Moletta, M. Welle, G. Alenyà, H. Yin, Z. Erickson, D. Held, J. Borràs, and D. Kragic. Unfolding the literature: A review of robotic cloth manipulation. *Annual Review of Control, Robotics, and Autonomous Systems*, 8: 295–322, 2025. doi:10.1146/annurev-control-022723-033252.

[3] J. Maitin-Shepard, M. Cusumano-Towner, J. Lei, and P. Abbeel. Cloth grasp point detection based on multiple-view geometric cues with application to robotic towel folding. In *IEEE International Conference on Robotics and Automation*, pages 2308–2315, 2010.

[4] M. Cusumano-Towner, A. Singh, S. Miller, J. F. O'Brien, and P. Abbeel. Bringing clothing into desired configurations with limited perception. In *IEEE International Conference on Robotics and Automation*, pages 3893–3900, 2011. doi:10.1109/ICRA.2011.5980327.

[5] I. Garcia-Camacho, M. Lippi, M. C. Welle, H. Yin, R. Antonova, A. Varava, J. Borràs, C. Torras, A. Marino, G. Alenyà, and D. Kragic. Benchmarking bimanual cloth manipulation. *IEEE Robotics and Automation Letters*, 5(2):1111–1118, 2020. doi:10.1109/LRA.2020.2965891.

[6] B. Jia, Z. Pan, Z. Hu, J. Pan, and D. Manocha. Cloth manipulation using random-forest-based imitation learning. *IEEE Robotics and Automation Letters*, 4(2):2086–2093, 2019. doi:10.1109/LRA.2019.2897370.

[7] J. Matas, S. James, and A. J. Davison. Sim-to-real reinforcement learning for deformable object manipulation. In *Proceedings of the 2nd Conference on Robot Learning*, volume 87 of *Proceedings of Machine Learning Research*, pages 734–743, 2018.

[8] D. Seita, A. Ganapathi, R. Hoque, M. Hwang, E. Cen, A. K. Tanwani, A. Balakrishna, B. Thananjeyan, J. Ichnowski, N. Jamali, K. Yamane, S. Iba, J. Canny, and K. Goldberg. Deep imitation learning of sequential fabric smoothing from an algorithmic supervisor. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2020.

[9] H. Ha and S. Song. Flingbot: The unreasonable effectiveness of dynamic manipulation for cloth unfolding. In *Proceedings of the 5th Conference on Robot Learning*, volume 164 of *Proceedings of Machine Learning Research*, pages 24–33, 2022.

[10] T. Weng, S. M. Bajracharya, Y. Wang, K. Agrawal, and D. Held. Fabricflownet: Bimanual cloth manipulation with a flow-based policy. In *Proceedings of the 5th Conference on Robot Learning*, volume 164 of *Proceedings of Machine Learning Research*, pages 192–202, 2022.

[11] Y. Avigal, L. Berscheid, T. Asfour, T. Kröger, and K. Goldberg. Speedfolding: Learning efficient bimanual folding of garments. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 1–8, 2022. doi:10.1109/IROS47612.2022.9981402.

[12] Z. Xu, C. Chi, B. Burchfiel, E. Cousineau, S. Feng, and S. Song. Dextairity: Deformable manipulation can be a breeze. In *Proceedings of Robotics: Science and Systems*, New York City, NY, USA, June 2022. doi:10.15607/RSS.2022.XVIII.017.

[13] X. Lin, Y. Wang, J. Olkin, and D. Held. Softgym: Benchmarking deep reinforcement learning for deformable object manipulation. In *Proceedings of the 2020 Conference on Robot Learning*, volume 155 of *Proceedings of Machine Learning Research*, pages 432–448, 2021.

[14] X. Lin, Y. Wang, Z. Huang, and D. Held. Learning visible connectivity dynamics for cloth smoothing. In *Proceedings of the 5th Conference on Robot Learning*, volume 164 of *Proceedings of Machine Learning Research*, pages 256–266, 2022.

[15] C. Chi and S. Song. Garmentnets: Category-level pose estimation for garments via canonical space shape completion. In *IEEE/CVF International Conference on Computer Vision*, pages 3324–3333, 2021.

[16] Z. Huang, X. Lin, and D. Held. Mesh-based dynamics with occlusion reasoning for cloth manipulation. In *Proceedings of Robotics: Science and Systems*, New York City, NY, USA, June 2022. doi:10.15607/RSS.2022.XVIII.011.

[17] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine. Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 1861–1870, 2018.

[18] S. Fujimoto, H. van Hoof, and D. Meger. Addressing function approximation error in actor-critic methods. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 1587–1596, 2018.

[19] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.

[20] A. Zeng, S. Song, S. Welker, J. Lee, A. Rodriguez, and T. Funkhouser. Learning synergies between pushing and grasping with self-supervised deep reinforcement learning. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 4238–4245, 2018. doi:10.1109/IROS.2018.8593986.

[21] P. Florence, L. Manuelli, and R. Tedrake. Dense object nets: Learning dense visual object descriptors by and for robotic manipulation. In *Proceedings of the 2nd Conference on Robot Learning*, volume 87 of *Proceedings of Machine Learning Research*, pages 373–385, 2018.

[22] A. Zeng, P. Florence, J. Tompson, S. Welker, J. Chien, M. Attarian, T. Armstrong, I. Krasin, D. Duong, V. Sindhwani, and J. Lee. Transporter networks: Rearranging the visual world for robotic manipulation. In *Proceedings of the 2020 Conference on Robot Learning*, volume 155 of *Proceedings of Machine Learning Research*, pages 726–747, 2021.

[23] L. Li, W. Chu, J. Langford, and R. E. Schapire. A contextual-bandit approach to personalized news article recommendation. In *Proceedings of the 19th International Conference on World Wide Web*, pages 661–670, 2010. doi:10.1145/1772690.1772758.

[24] P. Auer, N. Cesa-Bianchi, and P. Fischer. Finite-time analysis of the multiarmed bandit problem. *Machine Learning*, 47(2–3):235–256, 2002.

[25] N. Carion, L. Gustafson, Y.-T. Hu, S. Debnath, R. Hu, D. Suris, C. Ryali, K. V. Alwala, H. Khedr, A. Huang, J. Lei, T. Ma, B. Guo, A. Kalla, M. Marks, J. Greer, M. Wang, P. Sun, R. Rädle, T. Afouras, E. Mavroudi, K. Xu, T.-H. Wu, Y. Zhou, L. Momeni, R. Hazra, S. Ding, S. Vaze, F. Porcher, F. Li, S. Li, A. Kamath, H. K. Cheng, P. Dollár, N. Ravi, K. Saenko, P. Zhang, and C. Feichtenhofer. SAM 3: Segment Anything with Concepts, 2025. URL https://arxiv.org/abs/2511.16719.

A Hardware Setup

The transparent-table prototype uses two Intel RealSense D435 cameras: an overhead camera observing the cloth from above and an upward-facing camera mounted below the workspace to observe contact through the transparent support surface. The support surface is an off-the-shelf tempered-glass table that can be purchased commercially, rather than a custom optical bench. Dual-arm manipulation is performed with two Agile Nero robot arms positioned on opposite sides of the table. Figure 7 shows the physical layout.

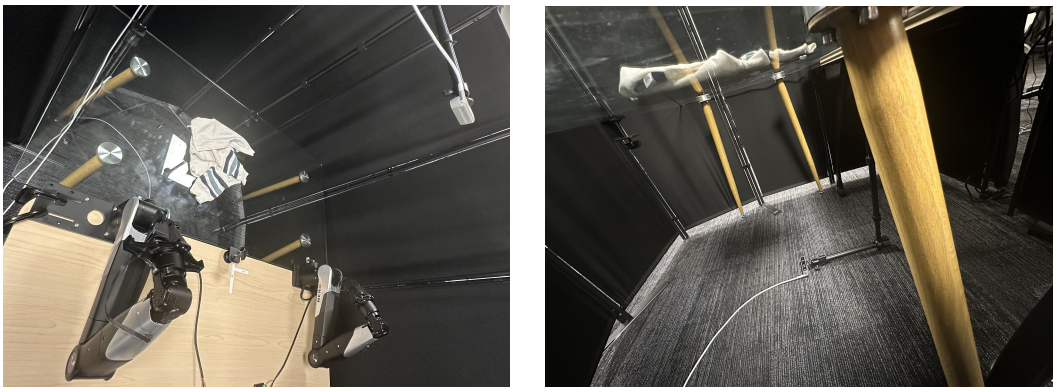


Figure 7: Physical transparent-table setup. Two Intel RealSense D435 cameras provide top and bottom observations through a commercially available tempered-glass table, while two Agile Nero arms perform dual-arm cloth manipulation.

B Qualitative Method Comparison Snapshots

The two policies in Section 4.3 are easiest to interpret side-by-side. Figure 8 shows deterministic rollouts of the top-only SAC baseline (top row) and the reward-bandit affordance policy (bottom row) starting from the identical crumpled cloth state used by every result in this paper. The two policies share the simulator seed and the same cloth-init snapshot, so any visible difference across the row is attributable to the policy and observation choice rather than to initial conditions. Cloth particles are coloured by height above the table; the red and blue spheres mark the left and right end-effectors.

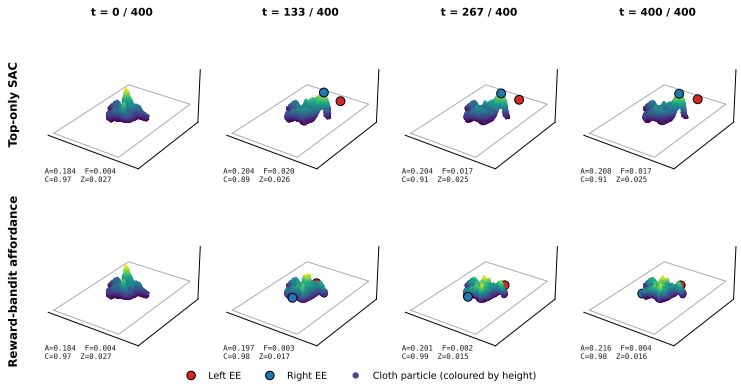


Figure 8: Deterministic dual-arm rollouts of the top-only SAC baseline (top row) and the reward-bandit affordance policy (bottom row) starting from the same crumpled cloth state. The reward-bandit affordance policy reaches a flatter, more grounded configuration with a folded residue roughly 4× smaller than the top-only baseline at the final frame.

Two qualitative differences are visible. First, the top-only policy never recovers the folded layer that sits along the front of the garment, while the reward-bandit affordance policy progressively flattens

it as the episode advances. Second, the top-only policy stalls with the right gripper held outside the cloth (a common failure mode that the residual frontier-target prior is designed to avoid). The per-frame area A, folded mask area F, contact ratio C, and height standard deviation Z printed under each snapshot quantify the gap: the reward-bandit affordance policy reaches $F \approx 0.004$ and $C \approx 0.98$ at $t{=}T$, versus $F \approx 0.017$ and $C \approx 0.91$ for the top-only baseline, which is consistent with the aggregated numbers in Table 1.

C Real-World Experimental Records

Figures 9 and 10 show eight real-world experimental records collected with the transparent-table setup across four garments. Figure 6 visualizes one representative trial with aligned progress metrics.

During execution, the system records synchronized RGB observations, robot states, segmentation outputs, and reward-related metrics. The top row of Figure 6 shows key moments from the execution sequence. The garment is segmented using SAM 3 [25], and the segmentation masks provide a direct visual indication of the observed garment state. Across the sequence, the robot progressively changes the garment configuration through physical interaction, producing a more favorable and observable cloth arrangement.

The bottom plot reports normalized task-relevant metrics extracted from the same record. To make the trend interpretable under real-world sensor noise, the plot keeps small local fluctuations while showing an uncertainty band. The plotted points are aligned with the keyframes above, so each image corresponds to the metric value at the same stage of execution. The overall upward trend suggests that the robot action sequence improves the measured garment state over time. This result indicates that the visual perception pipeline provides stable enough feedback for evaluating task progress despite occlusions, lighting variation, and cloth deformation.

D Implementation Details

The SAC implementation uses twin critics, automatic entropy tuning, replay updates after a random-action warmup, and deterministic evaluation rollouts for checkpoint selection. The auto-unfold preset selects checkpoints by deterministic evaluation: the long top/raw-dual runs use `eval_success_rate` with six evaluation episodes every 2000 updates, while the short scorer and occlusion studies use `eval_avg_return` with four evaluation episodes every 1000 updates. All final tables are produced from `latest_metrics.json` or from explicit post-hoc evaluation scripts, and incomplete runs are excluded by the finalizers.

The mask grid is 32×32 over a square table-centered ROI with half extent 0.40 m. Simulated SAM-style noise uses false-negative probability 0.01 and random flip probability 0.001. The bottom mask threshold is $\epsilon_{bot} = 0.035$ m above the table. The bottom mask is computed from cloth particles and table height, approximating an upward-looking transparent-table camera; it does not model calibration error, refraction/reflection, camera exposure, or learned segmentation bias.

Morphology is fixed unless stated otherwise. Both fold/contact frontier dilation and silhouette-edge erosion use one iteration of a 3×3 square binary kernel. The frontier is $\mathrm{dilate}(M^{fold})\,\mathrm{dilate}(M^{contact})M^{top}$. The residual prior uses pull distance 0.06 m in the reward-bandit runs, pick tolerance 0.045 m, prior weight $\lambda = 0.35$, and side-split candidate sets relative to the garment center. The initial linear scorer weights over [frontier, edge, fold, top, contact, outward] are $[4.0, 1.5, 1.0, 0.4, -0.2, 0.35]$; learned weights are clipped to $[0, 8] \times [0, 4] \times [0, 4] \times [0, 2] \times [-2, 2] \times [-1, 2]$. Reward-bandit runs use learning rate 0.20, exploration $\epsilon = 0.10$ over top-quartile candidate cells, baseline EMA coefficient 0.90, and weight decay 0.01 toward the heuristic initialization.

The visual encoders use feature dimension 128 and channel widths $(32, 64, 96)$. The top-mask encoder applies three convolutional blocks with stride 2 followed by global average pooling. The raw dual encoder builds separate top and bottom stems, tokenizes the final 8×8 feature maps into 64 tokens each, applies four-head top-to-bottom and bottom-to-top cross attention, and pools the

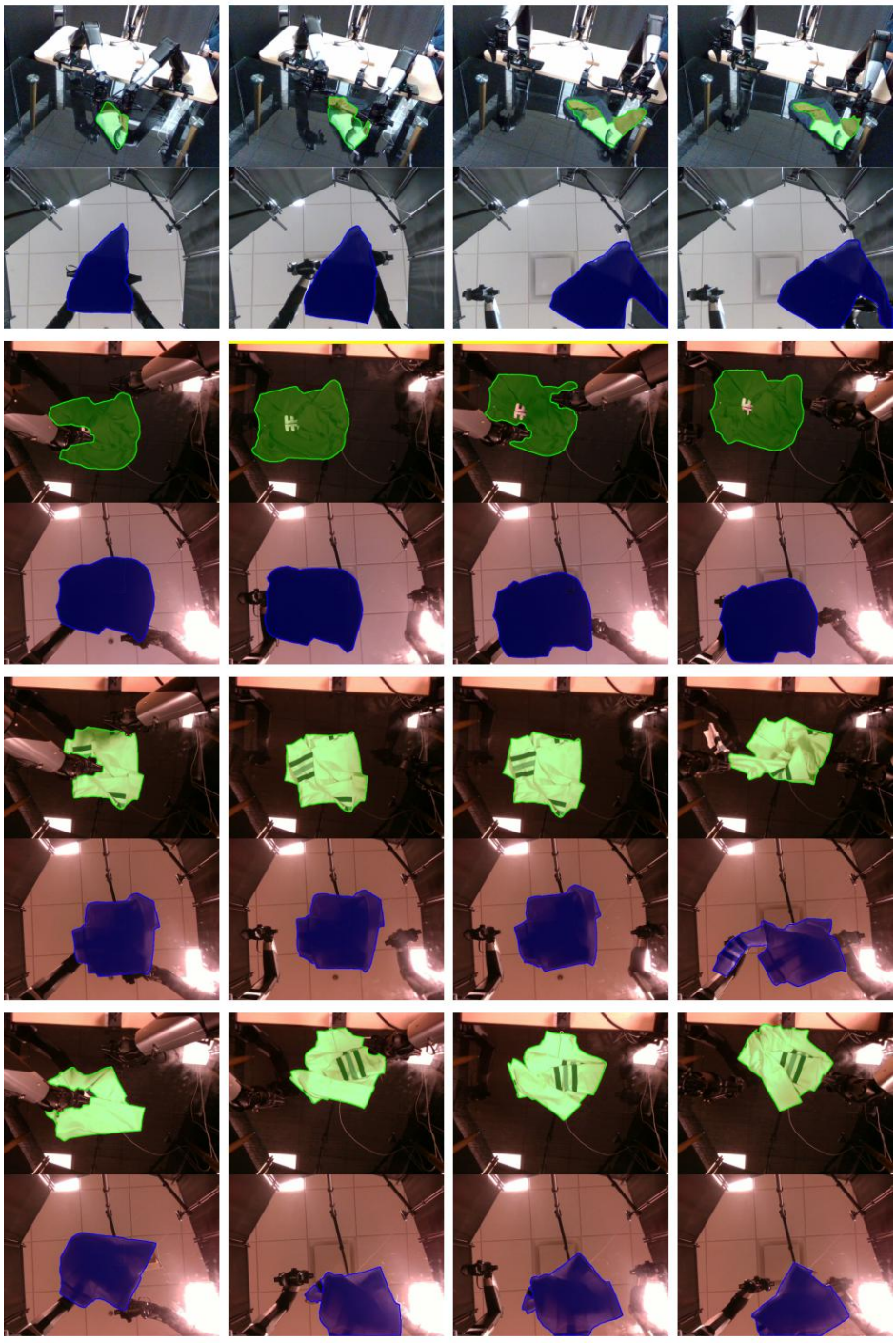


Figure 9: Real-world experimental records from the transparent-table setup (upper half). The full record contains eight trials across four garments. In each trial block, the top image is the top view and the bottom image is the bottom view.

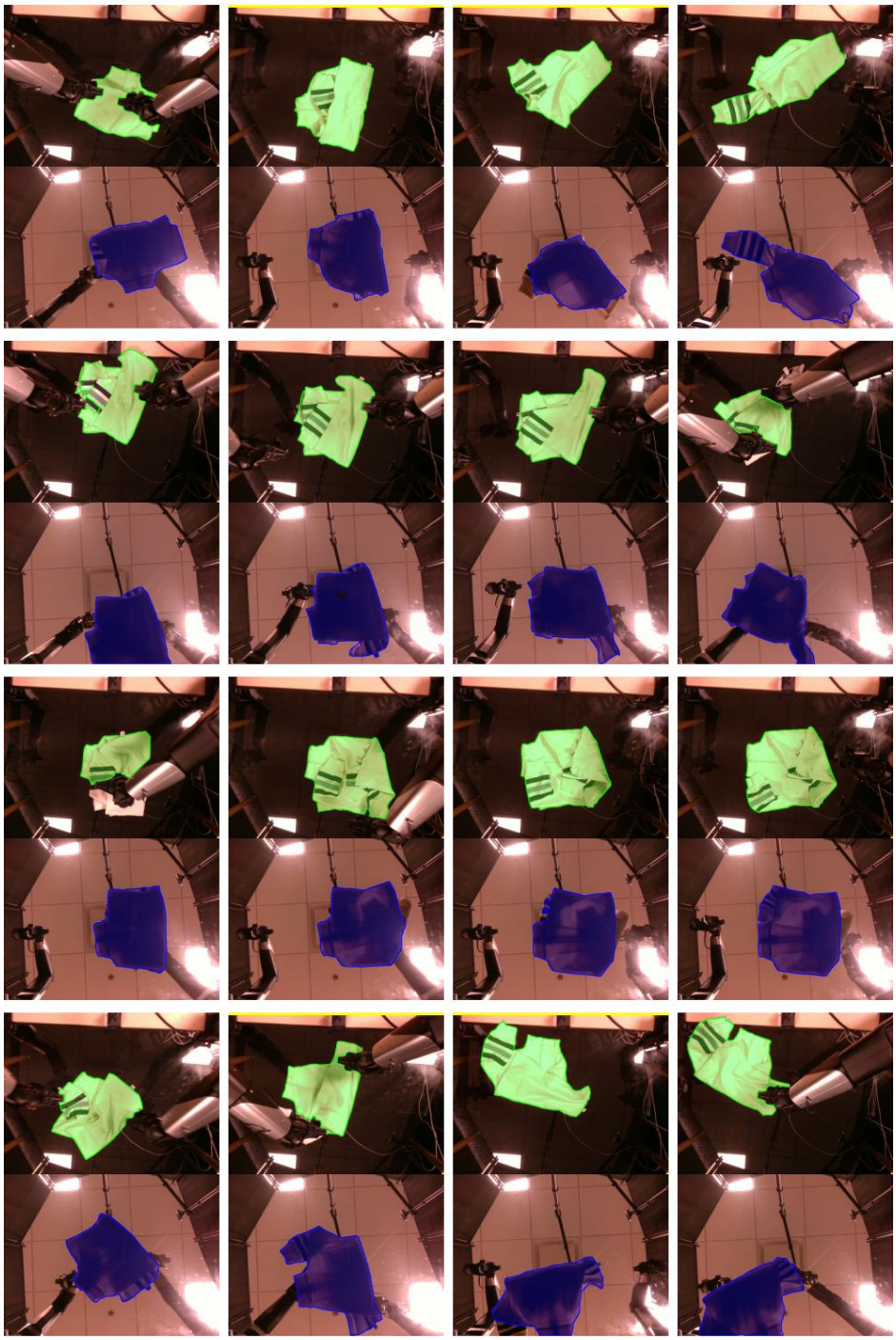


Figure 10: Real-world experimental records from the transparent-table setup (lower half), continued from Figure 9.

original and attended tokens. The affordance encoder uses a six-channel global stem, a two-channel fold/frontier stem, and a two-channel contact/edge stem, then applies the same bidirectional four-head attention and pooling before concatenating the visual feature with proprioception.

E Perception Validation

Figure 11 qualitatively validates the perception representation used by the real-world policy. Across different garment configurations, the bottom-view mask captures the cloth regions in contact with the transparent table, while the top-view mask captures the full visible garment silhouette. Combining these two views separates likely folded-over cloth (M_{fold}) from table-contacting cloth (M_{contact}). The resulting frontier mask consistently lies near visually salient fold boundaries, which are the regions where pulling is expected to expose hidden cloth and increase garment coverage. The selected pick points are sampled from high-scoring frontier regions and generally fall on or near these estimated fold-contact boundaries, supporting the use of bottom-camera contact perception as a practical signal for real-world cloth unfolding.

E.1 SAM3 Garment Segmentation Validation

To validate the robustness of the perception module across different garment appearances, we collected 12 real-world examples using the dual RealSense camera setup. For each example, the two RGB streams were vertically stitched and passed to SAM3 with the text prompt "cloth". We then applied a per-view post-processing step that keeps the highest-confidence denoised garment mask in each camera view. The resulting visualization is shown in Figure 12, where each row contains the raw stitched RGB observation, the SAM3 overlay, and the final binary mask.

Across the collected samples, SAM3 successfully detected the garment in both camera views for all 12 cases. The examples cover visually distinct clothing items, including dark, light, colored, patterned, and highly deformable garments. The detection confidence over the 24 retained per-view masks ranged from approximately 0.65 to 0.96, with an average score of about 0.85. These results indicate that the text-prompted SAM3 perception pipeline provides sufficiently stable garment masks for downstream real-world manipulation and paper-quality qualitative visualization.

F Long-Horizon Top vs. Raw Dual-View Runs

Table 3: One-million-step SAC runs used to evaluate whether raw bottom-view pixels add information beyond a top-only silhouette.

Condition	Obs.	Fusion	Episodes	Success	Return	Final area	Best area	Spread	Z-std
Top-mask	top_mask	top_cnn	2493	0.00	13.7 ± 2.0	0.200 ± 0.000	0.216 ± 0.000	0.183 ± 0.000	0.0271 ± 0.0003
Dual-view	dual_mask	cross_attention	2493	0.00	14.7 ± 1.8	0.202 ± 0.000	0.217 ± 0.000	0.184 ± 0.000	0.0218 ± 0.0001

G Action-Prior Probes

Table 4: No-learning 80-step bottom-view action-prior probe. Positive Δ area, Δ fold, Δ contact, and Δ z-std mean area increase, fold reduction, contact-ratio increase, and height-std reduction, respectively.

Condition	Score	Δ area	Δ fold	Δ contact	Δ z-std
Affordance prior	0.5414	0.0469	-0.0025	-0.0094	0.0059
Random dual	0.5237	0.0052	0.0025	0.0163	0.0071
Zero dual	0.5030	0.0072	0.0019	0.0120	0.0004
Affordance prior + jitter	0.4974	0.0204	-0.0069	-0.0312	0.0051

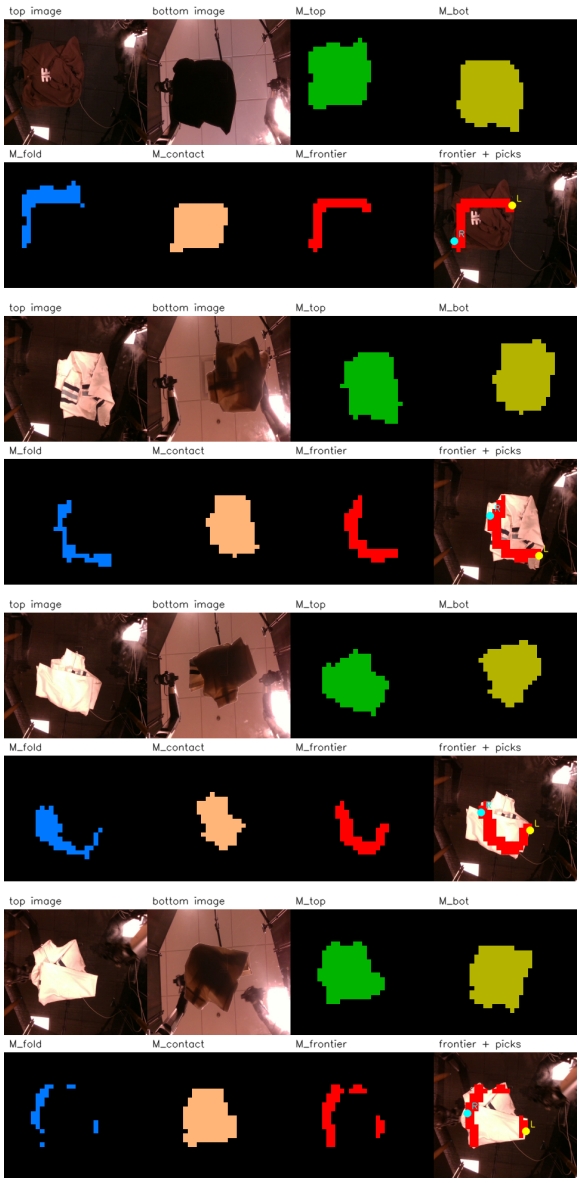


Figure 11: Perception validation for bottom-contact mask and fold-contact frontier in real-world scenes. Each row shows one recorded cloth state from the real robot setup. From left to right, the panels show the top RGB image, bottom RGB image through the transparent table, top cloth mask M_{top}, bottom contact mask M_{bot}, residual folded-cloth mask $M_{\text{fold}} = M_{\text{top}} - M_{\text{bot}}$, contact mask $M_{\text{contact}} = M_{\text{top}} \cap M_{\text{bot}}$, fold-contact frontier M_{frontier}, and the selected left/right pick points overlaid on the top image. The red frontier indicates the estimated boundary between upper folded cloth and lower table-contacting cloth. Yellow and cyan dots denote the selected grasp candidates for the two arms.

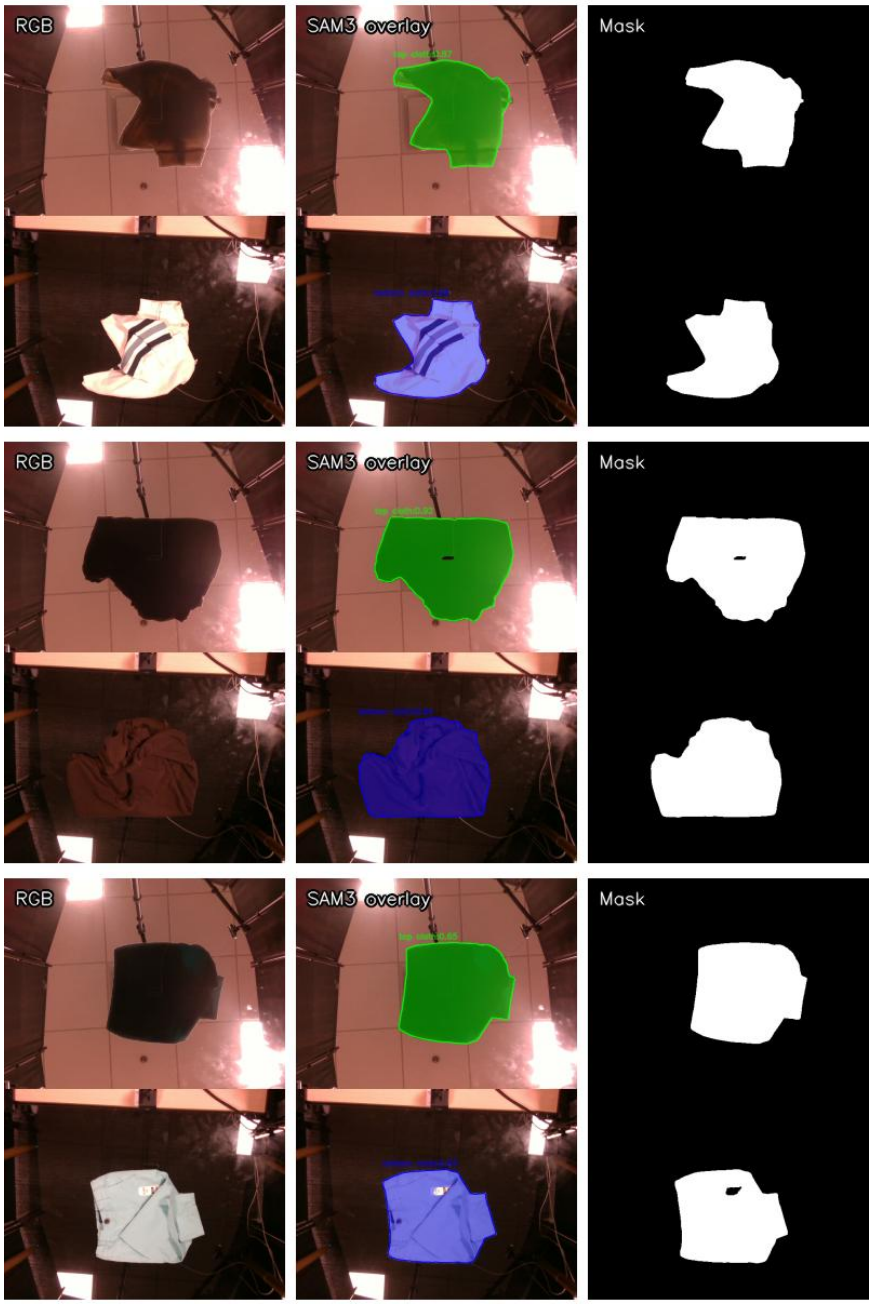


Figure 12: **Qualitative validation of SAM3 garment segmentation across real-world clothing instances.** Each row shows one captured garment configuration from the dual RealSense setup. The three columns show the stitched RGB observation, the SAM3 segmentation overlay, and the extracted binary garment mask, respectively. Using the text prompt "cloth", SAM3 consistently localizes the target garment in both camera views across 12 clothing appearances with different colors, textures, shapes, and degrees of deformation. This test evaluates bottom-view recognition of garments placed on a glass table. The per-view detections were filtered to retain one garment mask per camera view.

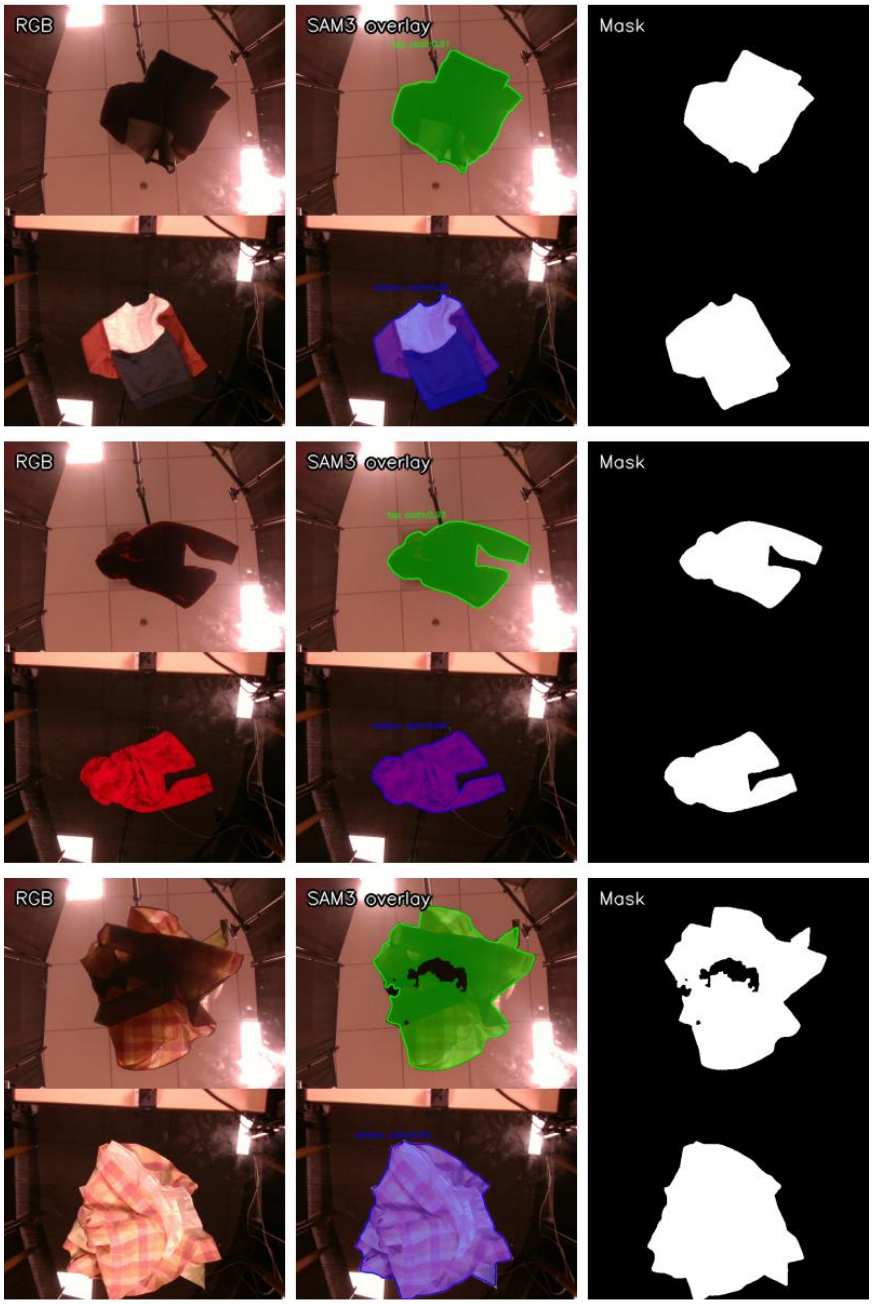


Figure 13: Qualitative validation of SAM3 garment segmentation across real-world clothing instances, continued from Figure 12.

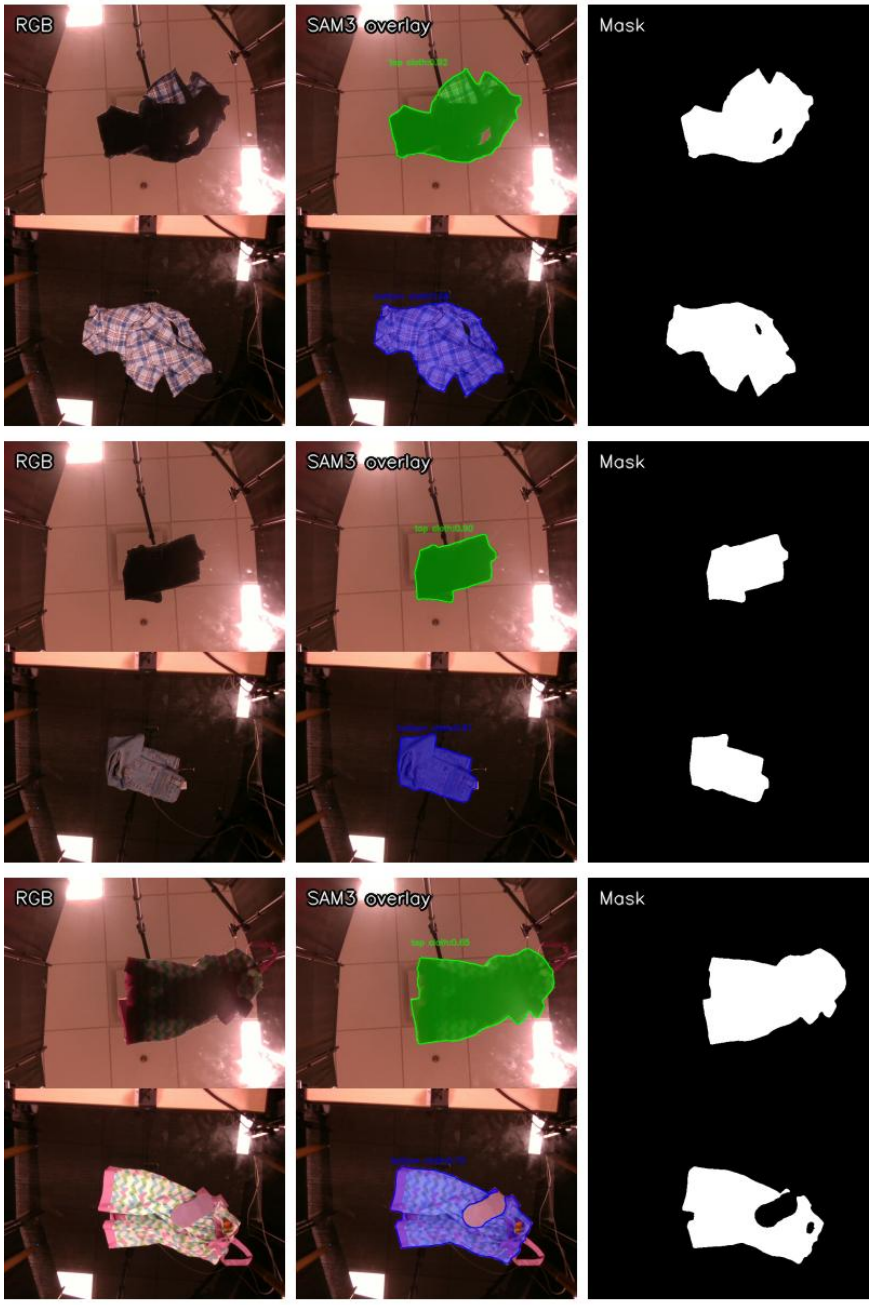


Figure 14: Qualitative validation of SAM3 garment segmentation across real-world clothing instances, continued from Figure 12.

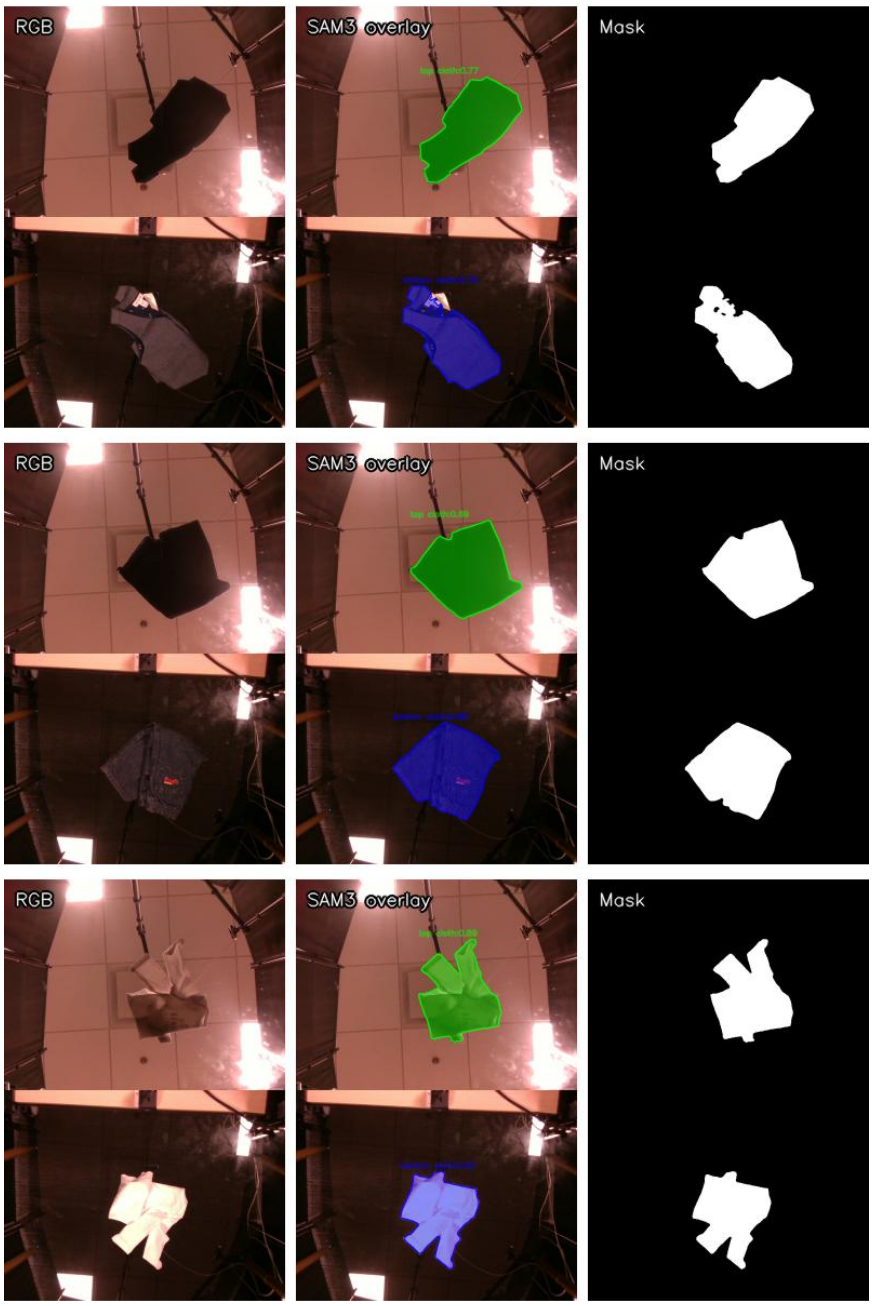


Figure 15: Qualitative validation of SAM3 garment segmentation across real-world clothing instances, continued from Figure 12.

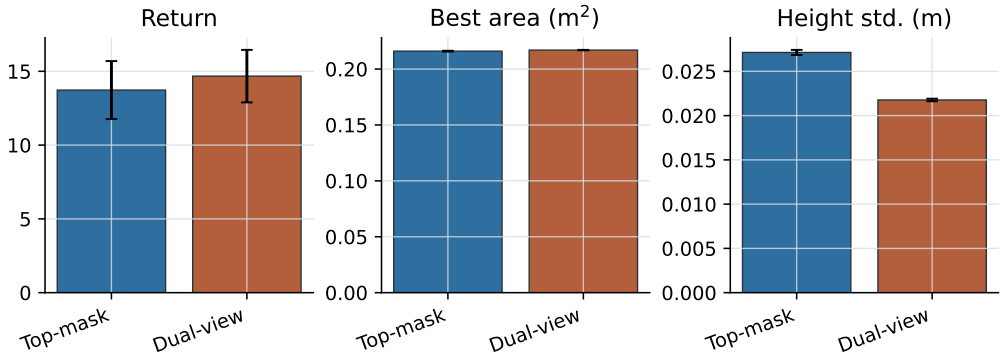


Figure 16: Aggregate episode statistics for the long top-mask and raw dual-view runs.

The prior is heuristic by construction. It depends on the binary morphology above, the side split, the pick tolerance, and the outward pull distance. Existing probes show that the prior can increase projected area without learning, but also that aggressive pulling can reduce contact or increase folding. We therefore treat the residual prior as a proposal generator for SAC, not as a complete controller.

Table 5: Lightweight sensitivity probe over outward pull distance, using two 80-step no-learning rollouts per setting. Negative Δ fold or Δ contact indicates that the prior increased folded area or reduced contact, respectively. Short pulls are safer than the default long pull.

Pull setting	Pull (m)	Score	Δ area	Δ fold	Δ contact	Final z-std
Short	0.06	0.521	0.0388	-0.0056	-0.0236	0.0181
Medium	0.09	0.509	0.0171	-0.0016	-0.0074	0.0182
Default	0.12	0.445	0.0063	-0.0138	-0.0724	0.0244

Table 6: Twelve-hour residual-prior comparison. These runs stopped before the requested 280k-step budget but completed about 248k steps and are used only as an exploratory diagnostic.

Variant	Steps	Score	Eval return	Area	Fold ↓	Contact ↑
Affordance residual prior	248598	0.650	18.7	0.199	0.0113	0.936
Affordance observation	248526	0.350	33.9	0.214	0.0141	0.924

H Discussion

The dual-view observation is designed to expose fold/contact information that is not identifiable from an overhead silhouette alone. The final interpretation should therefore focus on task-relevant geometry progress: final area, best area, spread, folded area, contact ratio, and height standard deviation. Binary success statistics can still be reported when a threshold is fixed, but they are secondary to these continuous measures because the threshold is task-dependent. Because the paper is compiled from run directories, any incomplete or failed experiment is excluded by the finalization script rather than silently reported.

I Robustness Diagnostics

Table 7 and Figure 17 give the full top-camera occlusion stress test behind the main-text summary. For non-top-only policies, bottom-view completion fills top-camera holes using the bottom contact mask. Relative to top-only SAC, the main gain is not z-std but better fold/contact and return. This stress

test supports robustness to top-camera occlusion, while not modeling bottom-camera calibration or optical failures.

Table 7: Structured top-camera occlusion stress test. "Arm-like" uses two elongated end-effector-centered occluders; bottom completion is enabled only for the reward-bandit affordance row.

Policy	Eval view	Score	Return	Area	Fold ↓	Z-std ↓
Top-only SAC	Clean	0.831 ± 0.027	27.8 ± 7.0	0.210 ± 0.007	0.0031 ± 0.0006	0.0204 ± 0.0011
Top-only SAC	Arm-like top occl.	0.810 ± 0.018	29.8 ± 3.3	0.210 ± 0.006	0.0058 ± 0.0022	0.0200 ± 0.0012
Raw dual-view SAC	Clean	0.692 ± 0.070	14.2 ± 4.9	0.155 ± 0.025	0.0021 ± 0.0008	0.0182 ± 0.0014
Raw dual-view SAC	Arm-like top occl.	0.708 ± 0.079	19.9 ± 2.1	0.181 ± 0.015	0.0073 ± 0.0036	0.0211 ± 0.0034
Reward-bandit no completion	Clean	0.819 ± 0.032	38.0 ± 3.4	0.222 ± 0.006	0.0094 ± 0.0034	0.0194 ± 0.0025
Reward-bandit no completion	Arm-like top occl.	0.827 ± 0.022	35.4 ± 3.8	0.213 ± 0.005	0.0038 ± 0.0010	0.0214 ± 0.0023
Reward-bandit affordance	Clean	0.809 ± 0.025	22.2 ± 10.7	0.210 ± 0.007	0.0072 ± 0.0036	0.0185 ± 0.0015
Reward-bandit affordance	Arm-like top occl. + bottom completion	0.860 ± 0.023	36.5 ± 2.9	0.220 ± 0.005	0.0033 ± 0.0008	0.0196 ± 0.0024
Raw dual occlusion-trained	Clean	0.701 ± 0.050	17.5 ± 4.6	0.186 ± 0.009	0.0075 ± 0.0017	0.0261 ± 0.0030
Reward-bandit occlusion-trained	Clean	0.802 ± 0.044	32.5 ± 2.2	0.216 ± 0.004	0.0092 ± 0.0045	0.0201 ± 0.0036
Top-only occlusion-trained	Clean	0.540 ± 0.049	23.3 ± 4.0	0.220 ± 0.005	0.0347 ± 0.0051	0.0406 ± 0.0044
Raw dual occlusion-trained	Arm-like top occl.	0.723 ± 0.046	18.3 ± 4.9	0.187 ± 0.010	0.0047 ± 0.0007	0.0259 ± 0.0033
Reward-bandit occlusion-trained	Arm-like top occl. + bottom completion	0.829 ± 0.022	32.8 ± 3.7	0.223 ± 0.005	0.0072 ± 0.0017	0.0220 ± 0.0020
Top-only occlusion-trained	Arm-like top occl.	0.563 ± 0.056	23.0 ± 3.6	0.221 ± 0.006	0.0324 ± 0.0058	0.0404 ± 0.0045

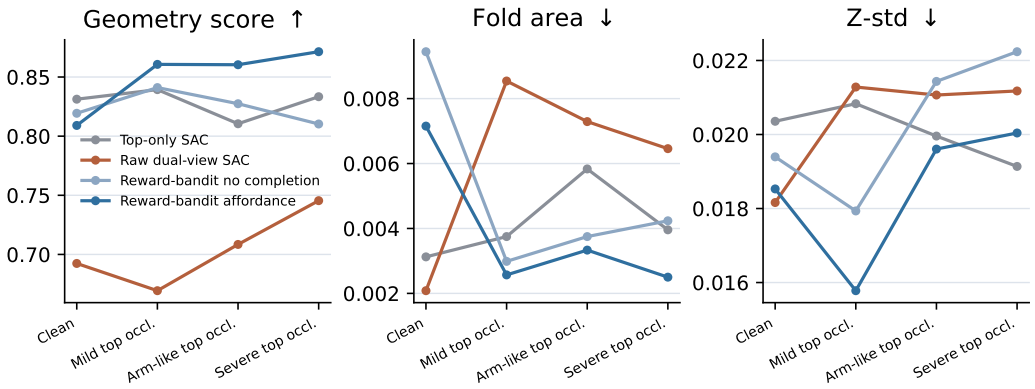


Figure 17: Occlusion stress test across increasing top-camera occluder sizes. The bottom-view reward-bandit affordance remains the strongest row under arm-like occlusion because the bottom contact mask supplies information not visible in the occluded top silhouette.

J Extended Validation Runs

The main paper keeps the core claims tied to completed one-million-step and 25k-step experiments. To address seed and horizon limitations, we also run a supplementary validation queue: three additional clean top-only/raw-dual seeds at 50k steps, and three reward-bandit versus three heuristic residual-prior seeds at 75k steps. Scores in Tables 8 and 9 are min-max normalized within each table only. At 50k steps over 3 top-only and 3 raw-dual seeds, raw dual changes z-std by 0.0263 relative to top-only and changes folded area by 0.0056. These runs are a mid-budget training-seed check rather than a replacement for a full multi-seed one-million-step study. At 75k steps over 3 reward-bandit seeds and 3 heuristic seeds, reward-bandit changes the within-table score by 0.168, z-std by -0.0021, and folded area by -0.0033 relative to the fixed scorer.

Table 8: Extended mid-budget top-only versus raw dual-view training-seed check. This table addresses whether the one-million-step pair is representative at shorter horizons; it does not replace full long-horizon multi-seed training.

Policy	Runs	Steps	Score	Return	Area	Fold ↓	Z-std ↓
Top-only SAC	3	50k	0.579 ± 0.173	27.0 ± 8.1	0.204 ± 0.005	0.0356 ± 0.0170	0.0418 ± 0.0143
Raw dual-view SAC	3	50k	0.445 ± 0.142	23.9 ± 6.2	0.194 ± 0.031	0.0412 ± 0.0051	0.0681 ± 0.0295

Table 9: Extended reward-bandit versus heuristic residual-prior check. The longer 75k-step horizon tests whether the 25k scorer conclusion remains visible after additional RL updates.

Policy	Runs	Steps	Score	Return	Area	Fold ↓	Z-std ↓
Heuristic residual prior	3	75k	0.506 ± 0.128	24.6 ± 2.6	0.203 ± 0.005	0.0121 ± 0.0045	0.0246 ± 0.0036
Reward-bandit residual prior	3	75k	0.675 ± 0.171	25.0 ± 3.5	0.206 ± 0.003	0.0088 ± 0.0031	0.0224 ± 0.0026

K Bottom-Mask Sensitivity Probe

To test sensitivity to ϵ_{bot}, morphology, side splitting, and bottom-mask failures without launching another full RL sweep, we collect 20 shared simulator states from three short crumpled-cloth rollouts and re-compute the affordance masks and heuristic pick targets offline. The baseline is $\epsilon_{bot} = 3.5$ cm with a one-iteration 3×3 square kernel and the table-centered side split used by the code. The table reports bottom-mask IoU and frontier IoU relative to this baseline, mean pick displacement for the two arms, the fraction of missing picks, and fold/contact area changes normalized by top-mask area. We extend the scenarios from the original submission to include $\epsilon_{bot} \in \{1.5, 2.0, 2.5, 4.5, 5.0\}$ cm, three percentile-based adaptive thresholds (p25, p35, p50), non-square dilation kernels ($1\times3, 3\times1, 1\times5, 5\times1, 5\times5, 7\times7$), a 3×3 kernel with two iterations, two side-split alternatives, and a broader noise family (5%/10% false negatives, 2% random flips, 1/2-pixel calibration shifts, block dropout, and a mixed failure).

The three takeaways are: *(i)* the offline frontier is largely insensitive to morphology and the precise ϵ_{bot} at and above 2.5 cm (frontier IoU ≥ 0.5, pick shift ≤ 5.4 cm); *(ii)* the choice of adaptive percentile in $\{25, 35, 50\}$ produces identical bottom and frontier masks on the shared states because the selected ϵ_{bot} is clipped to the same admissible interval; *(iii)* the dominant failure mode is calibration-style shift, which collapses frontier IoU to ≤ 0.07 and produces > 20 cm pick displacement even at one pixel. Policy-level evaluation of these perturbations is in Appendix L.

Table 10: Post-hoc sensitivity of bottom-view affordances and pick targets. This is a mask/proposal stability probe on shared states, not a policy retraining experiment. Low frontier IoU and large pick shift under shifted or mixed bottom-mask failures indicate that real calibration and segmentation quality are important assumptions.

Perturbation	Bottom IoU	Frontier IoU	Pick shift (cm)	Empty picks	Fold/contact Δ
$\epsilon_{bot} = 1.5$ cm	0.925	0.150	11.8	0.0%	0.074/-0.074
$\epsilon_{bot} = 2.0$ cm	0.967	0.307	9.3	0.0%	0.033/-0.033
$\epsilon_{bot} = 2.5$ cm	0.984	0.512	5.4	0.0%	0.015/-0.015
$\epsilon_{bot} = 4.5$ cm	0.993	0.668	2.7	0.0%	-0.007/0.007
$\epsilon_{bot} = 5.0$ cm	0.992	0.558	4.5	0.0%	-0.008/0.008
Adaptive p25	0.967	0.307	9.3	0.0%	0.033/-0.033
Adaptive p35	0.967	0.307	9.3	0.0%	0.033/-0.033
Adaptive p50	0.967	0.307	9.3	0.0%	0.033/-0.033
Kernel 1×3	1.000	0.536	0.5	0.0%	0.000/0.000
Kernel 3×1	1.000	0.403	0.8	0.0%	0.000/0.000
Kernel 1×5	1.000	0.489	0.2	0.0%	0.000/0.000
Kernel 5×1	1.000	0.401	0.6	0.0%	0.000/0.000
Kernel 5×5	1.000	0.508	0.2	0.0%	0.000/0.000
Kernel 7×7	1.000	0.344	0.4	0.0%	0.000/0.000
Kernel 3×3, 2 iters	1.000	0.508	0.2	0.0%	0.000/0.000
Top-centroid side split	1.000	1.000	0.0	0.0%	0.000/0.000
World-y side split	1.000	1.000	0.0	0.0%	0.000/0.000
Bottom 5% false negatives	0.949	0.123	10.2	0.0%	0.050/-0.050
Bottom 10% false negatives	0.897	0.078	12.5	0.0%	0.102/-0.102
Bottom 2% random flips	0.929	0.262	7.4	0.0%	0.020/-0.020
Bottom 1-pixel shift	0.758	0.065	23.2	0.0%	0.131/-0.131
Bottom 2-pixel shift	0.592	0.023	21.5	0.0%	0.248/-0.248
Bottom block dropout	0.951	0.566	4.1	0.0%	0.048/-0.048
Mixed bottom failure	0.752	0.061	11.2	0.0%	0.160/-0.160

L Bottom-Mask Noise Policy Evaluation

Table 2 reports the policy-level bottom-mask noise stress test in the main paper. For each base checkpoint we run six deterministic eval episodes per noise level, repeated across the three reward-bandit seeds; for raw-dual SAC we use the long 1M-step checkpoint; for top-only SAC we use the long 1M-step checkpoint and only the clean noise level because the top-only policy never consumes the bottom mask. The noise levels match the offline probe semantics so the appendix can be cross-referenced with Table 10. The noise-trained rows are produced by retraining the reward-bandit affordance policy with the same configurable perturbation family applied during training (3 seeds, 25k steps, "mixed" noise on every frame: FN 5%, flip 1%, shift 1 px), then evaluated with the same scripted noise grid.

Two observations support the main-text robustness summary. First, the dominant policy-level degradation tracks the offline ranking from Appendix K: 1-pixel shift and mixed failure produce the largest drops in geometry score, FN 5% and flip 2% are mild, and contact-coverage statistics (right-most column) move in the expected direction. Second, noise-augmented training recovers most of the score drop under the augmentation distribution without sacrificing clean-eval geometry, which is the standard sanity check for an augmentation-based defense. We do not claim that the noise-trained policy is robust to optical effects outside this configurable family, only that the in-distribution defense closes the gap.

M UCB-style Scorer Ablation

To address whether explicit uncertainty modeling improves the bandit-style update, we add a reward-UCB scorer variant that augments $s_t(i,j) = w_t^\top \phi_t(i,j)$ with a count-based exploration bonus $c\sqrt{\log(N{+}1)/(n_{ij}{+}1)}$ [24], where n_{ij} is the per-cell visit count and N the total picks since the last episode reset, and otherwise reuses the reward-bandit update rule, learning rate $\eta{=}0.20$, $\epsilon{=}0.10$ top-quartile exploration, and weight clipping. We run three matched seeds at 25k steps and report the result alongside reward-bandit in Table 11. The point of this ablation is not to introduce a new state-of-the-art scorer but to test whether replacing ϵ-greedy-on-top-quartile with an explicit visit-count bonus produces a measurable change at this scale; the rest of the pipeline is held fixed.

The two variants land in the same operating regime: reward-UCB has a slightly higher within-panel composite (0.607 vs 0.560), lower folded-mask area (0.0039 vs 0.0060), and modestly higher contact ratio (0.976 vs 0.968), but is essentially tied on coverage (0.203 vs 0.212 m^2) and z-std (0.0197 vs 0.0191 m), all within one standard error. The most visible difference is run-to-run stability: the reward-bandit return standard error is much larger (± 56.6 vs ± 4.6) because one of the three reward-bandit seeds collapsed during training, while no UCB seed did at this budget. This is consistent with the variance-control analysis in Appendix N.1: explicit visit-count uncertainty does not yield a measurable score improvement at the 25k-step scale used here, suggesting that the dominant variance source for this scorer family is feedback noise rather than under-exploration, but it can stabilize the failure mode of the basic ϵ-greedy variant.

Table 11: Reward-bandit vs. reward-UCB scorer ablation at matched 3 seeds and 25k SAC steps. The UCB variant adds a $c\sqrt{\log(N{+}1)/(n_{ij}{+}1)}$ exploration bonus to the same linear score; all other hyperparameters are identical. "Score" is the within-panel composite; absolute fold/contact/ z-std columns are the most informative comparison.

Variant	Runs	Score	Eval return	Area	Fold ↓	Contact ↑	Z-std ↓
Reward-bandit scorer	3	0.560 ± 0.230	-16.9 ± 56.6	0.212 ± 0.007	0.0060 ± 0.0018	0.968 ± 0.011	0.0191 ± 0.0031
Reward-UCB scorer	3	0.607 ± 0.188	29.9 ± 4.6	0.203 ± 0.010	0.0039 ± 0.0008	0.976 ± 0.007	0.0197 ± 0.0015

N Scorer Ablation Details

N.1 Scorer Update and Gating Pseudo-code

Algorithm M.1 gives the high-level policy-side logic summarized in Section 3. It intentionally omits implementation details such as empty-set checks and UCB counters; the following paragraph lists the variance controls used by the update.

Algorithm M.1 Reward-Bandit Scorer Update

initialize scorer weights at heuristic prior w_0 and EMA feedback baseline
for each environment step **do**
    build feasible per-arm target candidates from top-cloth cells
    rank candidates with the current affordance scorer
    fall back to the fixed heuristic if learned scores are uninformative
    select one target per arm with ϵ-greedy exploration
    execute the clipped residual prior action
    convert rollout signal to bounded, baseline-centered feedback
    update weights using centered target features, decay to w_0, and projection
end for

Variance control summary. Six mechanisms together keep the update stable in the experiments above: *(i)* bounded per-step feedback $p_t \in [-1, 1]$ via clipping or $\tanh$; *(ii)* EMA baseline $\bar{a}_t$ that removes the slow-moving component of p_t; *(iii)* clipped advantage $A_t \in [-1, 1]$; *(iv)* centered feature delta $\Delta\phi_t = \phi_t(u_t^r) - \bar{\phi}_t^r$ that controls the bias of the update direction; *(v)* weight decay $\rho(w_0 - w_t)$ that anchors the linear weights to the interpretable heuristic initialization; and *(vi)* hard per-coordinate projection $\Pi_{[l,u]}$. Together they guarantee $\|w_{t+1} - w_t\|_\infty \leq \eta + \eta\rho\|u - l\|_\infty$ for every step regardless of feedback realization.

N.2 Per-Seed Scorer Results

Figure 18 and Table 12 report the seed-level spread behind the 21-run scorer ablation summarized in Section 4.3. They are intended as a reproducibility check on the mean and standard-error results in Table 1, rather than as a separate claim. The scatter plot shows how the reward-bandit, progress-based, hybrid, and adaptive variants vary across seeds, while the table lists the corresponding per-seed evaluation metrics used to compute the main ablation summary.

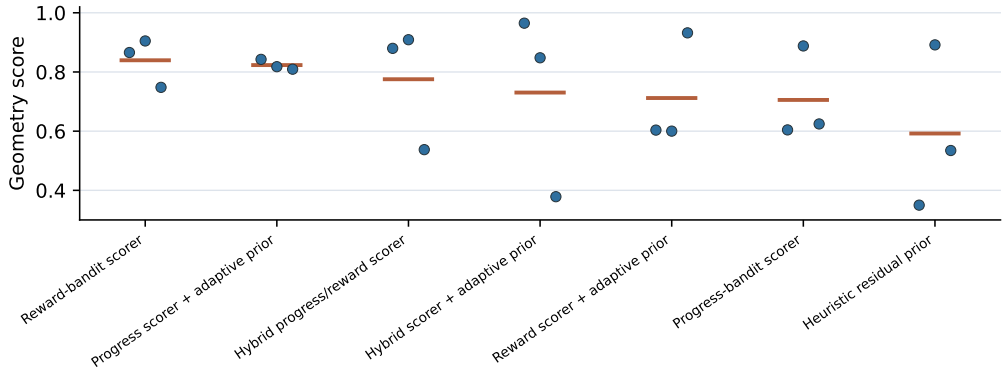


Figure 18: Per-seed geometry scores for the 21-run scorer ablation. Horizontal red segments denote condition means.

Table 12: Per-seed evaluation metrics for the scorer and adaptive-prior ablation.

Variant	Seed	Score	Return	Area	Fold ↓	Contact ↑
Reward-bandit scorer	940	0.866	30.6	0.207	0.0033	0.982
Reward-bandit scorer	941	0.905	32.7	0.213	0.0033	0.983
Reward-bandit scorer	942	0.748	37.0	0.220	0.0139	0.928
Progress scorer + adaptive prior	940	0.843	31.4	0.215	0.0072	0.965
Progress scorer + adaptive prior	941	0.818	31.3	0.210	0.0069	0.963
Progress scorer + adaptive prior	942	0.809	29.3	0.213	0.0077	0.960
Hybrid progress/reward scorer	940	0.880	32.2	0.214	0.0039	0.980
Hybrid progress/reward scorer	941	0.909	31.6	0.217	0.0033	0.983
Hybrid progress/reward scorer	942	0.538	17.4	0.179	0.0089	0.942
Hybrid scorer + adaptive prior	940	0.965	37.8	0.223	0.0030	0.985
Hybrid scorer + adaptive prior	941	0.848	37.0	0.220	0.0069	0.963
Hybrid scorer + adaptive prior	942	0.379	17.9	0.157	0.0094	0.925
Reward scorer + adaptive prior	940	0.604	27.5	0.190	0.0091	0.939
Reward scorer + adaptive prior	941	0.600	31.5	0.223	0.0227	0.886
Reward scorer + adaptive prior	942	0.932	37.8	0.225	0.0045	0.978
Progress-bandit scorer	940	0.604	30.3	0.227	0.0216	0.888
Progress-bandit scorer	941	0.888	29.4	0.208	0.0025	0.986
Progress-bandit scorer	942	0.624	28.0	0.220	0.0177	0.906
Heuristic residual prior	940	0.350	-68.0	0.229	0.0381	0.810
Heuristic residual prior	941	0.892	31.1	0.210	0.0034	0.981
Heuristic residual prior	942	0.535	23.0	0.210	0.0177	0.902